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1. Mainstream consumers driving plug-in battery-electric and plug-in hybrid electric cars: A qualitative analysis of responses and evaluations

January 2012
Ella Graham-Rowe | Benjamin Gardner | Charles Abraham | Stephen Skippon | Helga Dittmar | Rebecca Hutchins | Jenny Stannard

AbstractPlug-in electric vehicles can potentially emit substantially lower CO2 emissions than internal combustion engine vehicles, and so have the potential to reduce transport emissions without curtailing personal car use. Assessing the potential uptake of these new categories of vehicles requires an understanding of likely consumer responses. Previous in-depth explorations of appraisals and evaluations of electric vehicles have tended to focus on 'early adopters', who may not represent mainstream consumers. This paper reports a qualitative analysis of responses to electric cars, based on semi-structured interviews conducted with 40 UK non-commercial drivers (20 males, 20 females; age 24–70years) at the end of a seven-day period of using a battery electric car (20 participants) or a plug-in hybrid car (20 participants). Six core categories of response were identified: (1) cost minimisation; (2) vehicle confidence; (3) vehicle adaptation demands; (4) environmental beliefs; (5) impression management; and, underpinning all other categories, (6) the perception of electric cars generally as 'work in progress' products. Results highlight potential barriers to the uptake of current-generation (2010) plug-in electric cars by mainstream consumers. These include the prioritization of personal mobility needs over environmental benefits, concerns over the social desirability of electric vehicle use, and the expectation that rapid technological and infrastructural developments will make current models obsolete. Implications for the potential uptake of future electric vehicles are discussed.

2. Shopping online and/or in-store? A structural equation model of the relationships between e-shopping and in-store shopping

February 2007
Sendy Farag | Tim Schwanen | Martin Dijst | Jan Faber

AbstractSearching product information and buying goods online are becoming increasingly popular activities, which would seem likely to affect shopping trips. However, little empirical evidence about the relationships between e-shopping and in-store shopping is available. The aim of this study is to describe how the frequencies of online searching, online buying, and non-daily shopping trips relate to each other, and how they are influenced by such factors as attitudes, behaviour, and land use features. Questionnaire data were collected from 826 respondents residing in four municipalities (one urban, three suburban) in the centre of the Netherlands. Structural equation modelling was used to examine the variables' multiple and complex relationships. The results show that searching online positively affects the frequency of shopping trips, which in its turn positively influences buying online. An indirect positive effect of time-pressure on online buying was found and an indirect negative effect of online searching on shopping duration. These findings suggest that, for some people, e-shopping could be task-oriented (a time-saving strategy), and leisure-oriented for others. Urban residents shop online more often than suburban residents, because they tend to have a faster Internet connection. The more shopping opportunities one can reach within 10min by bicycle, the less often one searches online.

3. Is a new applied transportation research field emerging?––A review of intermodal rail–truck freight transport literature

January 2004
Y.M Bontekoning | C Macharis | J.J Trip

AbstractIntermodal freight transport has developed into a significant sector of the transport industry in its own right. This development has been followed by an increase in intermodal freight transportation research. We contend that a new transportation research application field is emerging; and that, while still in a pre-paradigmatic phase, it is now time to move on to a more mature independent research field. An independent research field can be justified because intermodal transport is a complex system that has characteristics which distinguishes it from other transport systems. We have reviewed 92 publications in order to identify the characteristics of the intermodal research community and scientific knowledge base. This paper will discuss aspects of this research, assessing the status quo and seeking directions for the future. To conclude, we will propose an intermodal research agenda which can direct the intermodal research field towards a period of "normal science".

4. Bicycling renaissance in North America? An update and re-appraisal of cycling trends and policies

July 2011
John Pucher | Ralph Buehler | Mark Seinen

AbstractThis paper reviews trends in cycling levels, safety, and policies in Canada and the USA over the past two decades. We analyze aggregate data for the two countries as well as city-specific case study data for nine large cities (Chicago, Minneapolis, Montréal, New York, Portland, San Francisco, Toronto, Vancouver, and Washington). Cycling levels have increased in both the USA and Canada, while cyclist fatalities have fallen. There is much spatial variation and socioeconomic inequality in cycling rates. The bike share of work commuters is more than twice as high in Canada as in the USA, and is higher in the western parts of both countries. Cycling is concentrated in central cities, especially near universities and in gentrified neighborhoods near the city center. Almost all the growth in cycling in the USA has been among men between 25–64years old, while cycling rates have remained steady among women and fallen sharply for children. Cycling rates have risen much faster in the nine case study cities than in their countries as a whole, at least doubling in all the cities since 1990. They have implemented a wide range of infrastructure and programs to promote cycling and increase cycling safety: expanded and improved bike lanes and paths, traffic calming, parking, bike-transit integration, bike sharing, training programs, and promotional events. We describe the specific accomplishments of the nine case study cities, focusing on each city's innovations and lessons for other cities trying to increase cycling. Portland's comprehensive package of cycling policies has succeeded in raising cycling levels 6-fold and provides an example that other North American cities can follow.

5. Value of travel time reliability: A review of current evidence

May 2012
Carlos Carrion | David Levinson

AbstractTravel time reliability is a fundamental factor in travel behavior. It represents the temporal uncertainty experienced by travelers in their movement between any two nodes in a network. The importance of the time reliability depends on the penalties incurred by the travelers. In road networks, travelers consider the existence of a trip travel time uncertainty in different choice situations (departure time, route, mode, and others). In this paper, a systematic review of the current state of research in travel time reliability, and more explicitly in the value of travel time reliability is presented. Moreover, a meta-analysis is performed in order to determine the reasons behind the discrepancy among the reliability estimates.

6. Customer segmentation revisited: The case of the airline industry

January 2008
Thorsten Teichert | Edlira Shehu | Iwan von Wartburg

AbstractAlthough the application of segmentation is a topic of central importance in marketing literature and practice, managers tend to rely on intuition and on traditional segmentation techniques based on socio-demographic variables. In the airline industry, it is regarded as common sense to separate between business and economy passengers. However, the simplicity of this segmentation logic no longer matches the ever more complex and heterogeneous choices made by customers. Airline companies relying solely on flight class as the segmentation criterion may not be able to customize their product offerings and marketing policies to an appropriate degree in order to respond to the shifting importance and growing complexity of customer choice drivers, e.g. flexibility and price as a result of liberalization in the airline industry. Thus, there is a need to re-evaluate the traditional market segmentation criterion.By analyzing the stated preference data of more than 5800 airline passengers, we show that segmenting into business and leisure (a) does not sufficiently capture the preference heterogeneity among customers and (b) leads to a misunderstanding of consumer preferences. We apply latent class modeling to our data and propose an alternative segmentation approach: we profile the identified segments along behavioral and socio-demographic variables. We combine our findings with observable consumer characteristics to derive pronounced fencing mechanisms for isolating and addressing customer segments receptive for tailored product packages.

7. Active transportation and physical activity: opportunities for collaboration on transportation and public health research

May 2004
James F Sallis | Lawrence D Frank | Brian E Saelens | M.Katherine Kraft

AbstractPhysically inactive lifestyles are a major public health challenge, and research in the transportation field on influences on the choice to walk and bike may provide guidance toward solutions. In the interests of promoting effective collaboration among the transportation, planning, and health fields, the current paper was written to fulfill three purposes. The first purpose was to summarize the transportation and planning

studies on the relation between community design and non-motorized ("active") transport and to interpret these studies from a health perspective. The second purpose was to summarize studies from the health literature that examine the relation between physical environmental variables and leisure-time physical activity that have relevance for transportation research. The third purpose was to promote more collaboration among transportation, planning, and health investigators by identifying opportunities for transdisciplinary research.

8. Integrating congestion pricing, transit subsidies and mode choice

July 2012
Leonardo J. Basso | Sergio R. Jara-Díaz

AbstractWe model and analyze optimal (welfare maximizing) prices and design of transport services in a bimodal context. Car congestion and transit design are simultaneously introduced and consumers choose based on the full price they perceive. The optimization variables are the congestion toll, the transit fare (and hence the level of subsidies) and transit frequency. We obtain six main results: (i) the optimal car-transit split is generally different from the total cost minimizing one; (ii) optimal congestion and transit price are interdependent and have an optimal frequency attached; (iii) the optimal money price difference together with the optimal frequency yield the optimal modal split; (iv) if this modal split is used in traditional stand-alone formulations – where each mode is priced independently–resulting congestion tolls and transit subsidies and fares are consistent with the optimal money price difference; (v) self-financing of the transport sector is feasible; and (vi) investment in car infrastructure induces an increase in generalized cost for all public transport users.

9. Can we reduce car use and, if so, how? A review of available evidence

June 2011
Ella Graham-Rowe | Stephen Skippon | Benjamin Gardner | Charles Abraham

AbstractTransport accounts for nearly a quarter of current energy-related carbon dioxide emissions with car travel constituting more than three quarters of all vehicle kilometres travelled. Interventions to change transport behaviour, and especially to reduce car use, could reduce CO2 emissions from road transport more quickly than technological measures. It is unclear, however, which interventions are effective in reducing car use and what the likely impact of these interventions would be on CO2 emissions. A two-stage systematic search was conducted focusing on reviews published since 2000 and primary intervention evaluations referenced therein. Sixty-nine reviews were considered and 47 primary evaluations found. These reported 77 intervention evaluations, including measures of car-use reduction. Evaluations of interventions varied widely in the methods they employed and the outcomes measures they reported. It was not possible to synthesise the findings using meta-analysis. Overall, the evidence base was found to be weak. Only 12 of the 77 evaluations were judged to be methodologically strong, and only half of these found that the intervention being evaluated reduced car use. A number of intervention approaches were identified as potentially effective but, given the small number of methodologically strong studies, it is difficult to draw robust conclusions from current evidence. More methodologically sound research is needed in this area.

10. Air transportation in a carbon constrained world: Long-term dynamics of policies and strategies for mitigating the carbon footprint of commercial aviation

December 2011
Sgouris Sgouridis | Philippe A. Bonnefoy | R. John Hansman

AbstractWith increasing demand for air transportation worldwide and decreasing marginal fuel efficiency improvements, the contribution of aviation to climate change relative to other sectors is projected to increase in the future. As a result, growing public and political pressures are likely to further target air transportation to reduce its greenhouse gas emissions. The key challenges faced by policy makers and air transportation industry stakeholders is to reduce aviation greenhouse gas emissions while sustaining mobility for passengers and time-sensitive cargo as well as meeting future demand for air transportation in developing and emerging countries. This paper examines five generic policies for reducing the emissions of commercial aviation; (1) technological efficiency improvements, (2) operational efficiency improvements, (3) use of alternative fuels, (4) demand shift and (5) carbon pricing (i.e. market-based incentives). In order to evaluate the impacts of these policies on total emissions, air transport mobility, airfares and airline profitability, a system dynamics modeling approach was used. The Global Aviation Industry Dynamics (GAID) model captures the systemic interactions and the delayed feedbacks in the air transportation system and allows scenarios testing through simulations. For this analysis, a set of 34 scenarios with various levels of aggressiveness along the five generic policies were simulated and tested. It was found that no single policy can maintain emissions levels steady while increasing projected demand for air transportation. Simulation results suggest that a combination of the proposed policies does produce results that are close to a "weak" sustainability definition of increasing supply to meet new demand needs while maintaining constant or increasing slightly emissions levels. A combination of policies that includes aggressive levels of technological and operations efficiency improvements, use of biofuels along with moderate levels of carbon

pricing and short-haul demand shifts efforts achieves a 140% increase in capacity in 2024 over 2004 while only increasing emissions by 20% over 2004. In addition, airline profitability is moderately impacted (10% reduction) compared to other scenarios where profitability is reduced by over 50% which pose a threat to necessary investments and the implementation of mitigating measures to reduce CO2 emissions. This study has shown that an approach based on a portfolio of mitigating measures and policies spanning across technology and operational improvements, use of biofuels, demand shift and carbon pricing is required to transition the air transportation industry close to an operating point of environmental and mobility sustainability.

11. How derived is the demand for travel? Some conceptual and measurement considerations

September 2001
Patricia L. Mokhtarian | Ilan Salomon

AbstractThis paper contests the conventional wisdom that travel is a derived demand, at least as an absolute. Rather, we suggest that under some circumstances, travel is desired for its own sake. We discuss the phenomenon of undirected travel – cases in which travel is not a byproduct of the activity but itself constitutes the activity. The same reasons why people enjoy undirected travel (a sense of speed, motion, control, enjoyment of beauty) may motivate them to undertake excess travel even in the context of mandatory or maintenance trips. One characteristic of undirected travel is that the destination is ancillary to the travel rather than the converse which is usually assumed. We argue that the destination may be to some degree ancillary more often than is realized. Measuring a positive affinity for travel is complex: in self-reports of attitudes toward travel, respondents are likely to confound their utility for the activities conducted at the destination, and for activities conducted while traveling, with their utility for traveling itself. Despite this measurement challenge, preliminary empirical results from a study of more than 1900 residents of the San Francisco Bay Area provide suggestive evidence for a positive utility for travel, and for a desired travel time budget (TTB). The issues raised here have clear policy implications: the way people will react to policies intended to reduce vehicle travel will depend in part on the relative weights they assign to the three components of a utility for travel. Improving our forecasts of travel behavior may require viewing travel literally as a “good” as well as a “bad” (disutility).

12. Evaluating the impact of legislation prohibiting hand-held cell phone use while driving

March 2010
Alexander G. Nikolaev | Matthew J. Robbins | Sheldon H. Jacobson

AbstractAs of November 2008, the number of cell phone subscribers in the US exceeded 267 million, nearly three times more than the 97 million subscribers in June 2000. This rapid growth in cell phone use has led to concerns regarding their impact on driver performance and road safety. Numerous legislative efforts are under way to restrict hand-held cell phone use while driving. Since 1999, every state has considered such legislation, but few have passed primary enforcement laws. As of 2008, six states, the District of Columbia (DC), and the Virgin Islands have laws banning the use of hand-held cell phones while driving. A review of the literature suggests that in laboratory settings, hand-held cell phone use impairs driver performance by increasing tension, delaying reaction time, and decreasing awareness. However, there exists insufficient evidence to prove that hand-held cell phone use increases automobile-accident-risk. In contrast to other research in this area that uses questionnaires, tests, and simulators, this study analyzes the impact of hand-held cell phone use on driving safety based on historical automobile-accident-risk-related data and statistics, which would be of interest to transportation policy-makers. To this end, a pre-law and post-law comparison of automobile accident rate measures provides one way to assess the effect of hand-held cell phone bans on driving safety; this paper provides such an analysis using public domain data sources. A discussion of what additional data are required to build convincing arguments in support of or against legislation is also provided.

13. Scientific research about climate change mitigation in transport: A critical review

December 2011
Tim Schwanen | David Banister | Jillian Anable

AbstractThis paper seeks to develop a deeper understanding of the research on climate change mitigation in transport. We suggest that work to date has focused on the effects of improvements in transport technologies, changes in the price of transport, physical infrastructure provision, behavioural change and alternative institutional arrangements for governing transport systems. In terms of research methodologies, positivist and quantitative analysis prevails, although there are signs of experimentation with non-positivist epistemologies and participatory methods. These particular engagements with climate change mitigation reflect mutually reinforcing tendencies within and beyond the academic transport community. We first draw on a revised version of Thomas Kuhn’s philosophy of science to explore the path dependencies within transport studies, which are at least partly responsible for the predisposition towards quantitative modelling and technology, pricing and infrastructure oriented interventions in transport systems. We then employ the governmentality perspective to examine how transport academics’ engagements with climate change

mitigation depend on and align with more general understandings of climate change in UK society and beyond. The analysis makes clear that ecological modernisation and neo-liberal governmentality more generally provide the context for the current focus on and belief in technological, behaviour change, and especially market-based mitigation strategies. While current research trajectories are important and insightful, we believe that a deeper engagement with theoretical insights from the social sciences will produce richer understandings of transport mitigation in transport and briefly outline some of the contributions thinking on socio-technical transitions and practice theories can make.

14. Urban density and energy consumption: a new look at old statistics

February 2004
Orit Mindali | Adi Raveh | Ilan Salomon

AbstractThere is growing concern about the negative environmental and energy effects caused by transportation systems and related land-use patterns. Travel and land-use are a function of one another, therefore it is often hypothesized that changing urban structure can result in changes in energy consumption. A popular view suggests that there is a strong negative correlation between urban density and energy consumption. This implies that increasing density will result in a reduction in energy consumption [Cities and Automobile Dependence: An International Sourcebook, Avebury Technical, Great Britain, 1989].Using Co-Plot, an innovative multivariate statistical technique, this research crystallizes some of the relationships between density and energy consumption in western cities. The method is applied on Newman and Kenworthy's data, leading to the conclusion that there is no direct impact of total urban density. Instead several other relationships between energy consumption and density attributes can be identified.

15. The effect of the price of gasoline on the urban economy: From route choice to general equilibrium

July 2012
Alex Anas | Tomoru Hiramatsu

AbstractRELU-TRAN2, a spatial computable general equilibrium (CGE) model of the Chicago MSA is used to understand how gasoline use, car-VMT, on-the-road fuel intensity, trips and location patterns, housing, labor and product markets respond to a gas price increase. We find a long-run elasticity of gasoline demand (with congestion endogenous) of −0.081, keeping constant car prices and the TFI (technological fuel intensity) of car types but allowing consumers to choose from car types. 43% of this long run elasticity is from switching to transit; 15% from trip, car-type and location choice; 38% from price, wage and rent equilibration, and 4% from building stock changes. 79% of the long run elasticity is from changes in car-VMT (the extensive margin) and 21% from savings in gasoline per mile (the intensive margin); with 83% of this intensive margin from changes in congestion and 17% from the substitution in favor of lower TFI. An exogenous trend-line improvement of the TFI of the car-types available for choice raises the long-run response to a percent increase in the gas price from −0.081 to −0.251. Thus, only 1/3 of the long-run response to the gas price stems from consumer choices and 2/3 from progress in fuel intensity. From 2000 to 2007, real gas prices rose 53.7%, the average car fuel intensity improved 2.7% and car prices fell 20%. The model predicts that from these changes alone, keeping constant population, income, etc. aggregate gasoline use in this period would have fallen by 5.2%.

16. Car use: lust and must. Instrumental, symbolic and affective motives for car use

February–March 2005
Linda Steg

AbstractThis paper reports results of two questionnaire studies aimed at examining various motives for car use. In the first study, a random selection of 185 respondents who possess a driving licence were interviewed. Respondents were recruited from the cities of Groningen and Rotterdam, The Netherlands. The sample of the second study comprised a random selection of 113 commuters who regularly travelled during rush hours in and around Rotterdam, a region in the west of the Netherlands. First, it was examined which categories of car use motives may be distinguished. As proposed by Dittmar's (1992) [The social psychology of material possessions: to have is to be. Havester Wheatsheaf, Hemel Hempstead, UK; St. Martin's Press, New York] model on the meaning of material possessions, results from both studies revealed that car use not only fulfils instrumental functions, but also important symbolic and affective functions. Second, it was studied to what extent these different motives are related to the level of car use. From the results of study 2, it appeared that commuter car use was most strongly related to symbolic and affective motives, and not to instrumental motives. Third, individual differences in the relative importance of the three categories of motives were investigated. In both studies, most group differences were found in the evaluation of the symbolic and affective motives (and not the instrumental ones). Especially frequent drivers, respondents with a positive car attitude, male and younger respondents valued these non-instrumental motives for car use. These results suggest that policy makers should not exclusively focus on instrumental motives for car use, but they should consider the many social and affective motives as well.

17. Performance indicators for public transit connectivity in multi-modal transportation networks

August 2012
Sabyasachee Mishra | Timothy F. Welch | Manoj K. Jha

AbstractConnectivity plays a crucial role as agencies at the federal and state level focus on expanding the public transit system to meet the demands of a multimodal transportation system. Transit agencies have a need to explore mechanisms to improve connectivity by improving transit service. This requires a systemic approach to develop measures that can prioritize the allocation of funding to locations that provide greater connectivity, or in some cases direct funding towards underperforming areas. The concept of connectivity is well documented in social network literature and to some extent, transportation engineering literature. However, connectivity measures have limited capability to analyze multi-modal public transportation systems which are much more complex in nature than highway networks.In this paper, we propose measures to determine connectivity from a graph theoretical approach for all levels of transit service coverage integrating routes, schedules, socio-economic, demographic and spatial activity patterns. The objective of using connectivity as an indicator is to quantify and evaluate transit service in terms of prioritizing transit locations for funding; providing service delivery strategies, especially for areas with large multi-jurisdictional, multi-modal transit networks; providing an indicator of multi-level transit capacity for planning purposes; assessing the effectiveness and efficiency for node/stop prioritization; and making a user friendly tool to determine locations with highest connectivity while choosing transit as a mode of travel. An example problem shows how the graph theoretical approach can be used as a tool to incorporate transit specific variables in the indicator formulations and compares the advantage of the proposed approach compared to its previous counterparts. Then the proposed framework is applied to the comprehensive transit network in the Washington–Baltimore region. The proposed analysis offers reliable indicators that can be used as tools for determining the transit connectivity of a multimodal transportation network.

18. Residential location and transit-oriented development in a new rail corridor

March 2011
Doina Olaru | Brett Smith | John H.E. Taplin

AbstractThe relationship of form, use, and density in urban development and their influence on human behavior and travel is a key element of many land use and transport policies. Prior research indicates high-density urban development leads to decreased travel and thus sustainable mobility; however, personal attitudes seem to have greater effect on mobility than does the urban form. This research evaluates how households consider transit-oriented development (TOD) characteristics in their location decisions with regard to new Mandurah railway line stations opened in December 2007 in Perth, Western Australia. The results indicate that the choice of residence reflects neighborhood and housing attributes, with significant heterogeneity in the populations of the three precincts in terms of their valuation of various housing characteristics, proximity to urban facilities, and transport. There is also significant variation in households' attitudes to natural and artificial environments. A better understanding of the complex relationships among environment, travel, socio-demographic characteristics, and household attitudes can help transport planners leverage the benefits of TOD and improve the quality of urban design and community life.

19. Transit network design and scheduling: A global review

December 2008
Valérie Guihaire | Jin-Kao Hao

AbstractThis paper presents a global review of the crucial strategic and tactical steps of transit planning: the design and scheduling of the network. These steps influence directly the quality of service through coverage and directness concerns but also the economic profitability of the system since operational costs are highly dependent on the network structure. We first exhibit the context and the goals of strategic and tactical transit planning. We then establish a terminology proposal in order to name sub-problems and thereby structure the review. Then, we propose a classification of 69 approaches dealing with the design, frequencies setting, timetabling of transit lines and their combinations. We provide a descriptive analysis of each work so as to highlight their main characteristics in the frame of a two-fold classification referencing both the problem tackled and the solution method used. Finally, we expose recent context evolutions and identify some trends for future research. This paper aims to contribute to unification of the field and constitutes a useful complement to the few existing reviews.

20. Poverty, policy, and public transportation

June 2008
Thomas W. Sanchez

AbstractBy all appearances, the circumstances surrounding employment and income distribution in the United States have remained notably the same over the past 30–40 years. At the same time, policies for improving the conditions of low-income persons have remained relatively unchanged. Relevant published accounts continue to cite poorly integrated residential and employment location patterns and poor public

transportation service as critical obstacles to improving the economic and social conditions of low-income persons. The relationship between poverty and public transportation was researched extensively during the late 1960s and the early 1970s; however, little recognition has been given to these efforts by more recent research efforts. To learn from the past we should review public transportation policies from 1960 to 2000 to highlight federal policies that affected urban areas during this time period, especially in relation to low-income transportation mobility.

21. Urban spatial location advantage: The dual of the transportation problem and its implications for land-use and transport planning

January 2012
Enda Murphy

AbstractNumerous recent studies have investigated the relationship between the location of jobs and housing in urban areas and how this relates to urban commuting patterns. Few have utilised the dual of the transportation problem of linear programming (TPLP) to provide insights into these relationships Accordingly, this analysis utilises the TPLP to determine dual variable values (shadow prices) for a study area in Dublin, Ireland. The approach determines the pattern of relative location advantage for the peak and off-peak travel periods and for public and private transport for 1991 and 2001. The results are set against the expected results for hypothetical urban structures. The results show that the pattern of relative location advantage has altered sharply over the study period for off-peak trip-making but has remained more or less the same for trip-making in the peak period. For the off-peak period, the pattern of relative location advantage has shifted from the central area to the periphery specifically for private transport trips; for public transport, the pattern has remained focused on the city centre. This indicates that private transport users can react more quickly to changes in the distribution of land-use activities than their public transport counterparts due to the relatively fixed nature of the latter mode. This implies that the public transport network needs to be reorganized to better reflect the revised pattern of trip-making specifically for the off-peak period. The results demonstrate the value of using the approach for providing information about the spatial organisation of land uses within cities and where future development may be targeted.

22. Road network vulnerability analysis of area-covering disruptions: A grid-based approach with case study

June 2012
Erik Jenelius | Lars-Göran Mattsson

AbstractWe present an approach to systematically analysing the vulnerability of road networks under disruptions covering extended areas. Since various kinds of events including floods, heavy snowfall, storms and wildfires can cause such spatially spread degradations, the analysis method is an important complement to the existing studies of single link failures. The methodology involves covering the study area with grids of uniformly shaped and sized cells, where each cell represents the extent of an event disrupting any intersecting links. We apply the approach to the Swedish road network using travel demand and network data from the Swedish national transport modelling system Sampers. The study shows that the impacts of area-covering disruptions are largely determined by the level of internal, outbound and inbound travel demand of the affected area itself. This is unlike single link failures, where the link flow and the redundancy in the surrounding network determine the impacts. As a result, the vulnerability to spatially spread events shows a markedly different geographical distribution. These findings, which should be universal for most road networks of similar scale, are important in the planning process of resource allocation for mitigation and recovery.

23. Much Ado about Nothing? – An analysis of economic impacts and ecologic effects of the EU-emission trading scheme in the aviation industry

December 2011
Jan Vespermann | Andreas Wald

AbstractFrom 2012 on, all CO_2 emissions from flights departing from or arriving at airports within the European Union have to be offset. We analyze the economic and ecological impacts that are caused by an inclusion of the aviation industry into the proposed emissions trading scheme (ETS). Building on the now fixed system design we employ a simulation model to estimate the impacts of the scheme. Our results indicate that financial impacts are highly dependant on external settings, such as allowance prices and demand growth. We show that the financial burden on the aviation industry will be rather modest in the first years after the introduction of the system and therefore induce only low competition distortions. Likewise, emission reductions within air transportation will be comparably low. While aviation will induce a decline of emissions in other sectors, significant absolute reductions within air transportation can only be reached by a more restrictive system design.

24. Contemplating cycling to work: Attitudes and perceptions in different stages of change

May 2007
Birgitta Gatersleben | Katherine M. Appleton

AbstractIn 1996 the UK government introduced the National Cycle Strategy which aimed to double the number of cycling trips by the end of 2002 and double them again by 2012. So far, however, these targets have not been met. The House of Commons ascribes this to 'a fundamental lack of commitment to cycling on an individual, regional and national level'. This paper addresses the individual level by examining the views of commuters in different stages of change as distinguished by Prochaska's model [Prochaska, J.O., DiClemente, C.C., 1984. The Transtheoretical Approach: Crossing Traditional Boundaries of Change. Dow Jones/Irwin, Homewood IL]. This model views behaviour change as a process rather than an event. Two studies were conducted amongst university staff and students: a survey study and an action study. The studies showed that as people progress from precontemplation to action their attitudes towards cycling become more positive and their perceptions of various personal and external barriers change. This suggests that different strategies are necessary to move people in different stages of change to action and maintenance. At the moment, it seems that regular cyclists form a very small minority of people who will cycle under most circumstances simply because they like cycling. The majority of people have never contemplated cycling. There is, however, also a group of people who would like to cycle and could be persuaded to cycle under the right circumstances.

25. The technical efficiency of container ports: Comparing data envelopment analysis and stochastic frontier analysis

May 2006
Kevin Cullinane | Teng-Fei Wang | Dong-Wook Song | Ping Ji

AbstractThe efficiency of the container port industry has been variously studied utilising either Data Envelopment Analysis (DEA) or Stochastic Frontier Analysis (SFA). Given the strengths and weaknesses associated with these two approaches, the efficiency estimates and scale properties derived from these analyses are not always convincing. This paper applies both approaches to the same set of container port data for the world's largest container ports and compares the results obtained. A high degree of correlation is found between the efficiency estimates derived from all the models applied, suggesting that results are relatively robust to the DEA models applied or the distributional assumptions under SFA. High levels of technical efficiency are associated with scale, greater private-sector participation and with transhipment as opposed to gateway ports. In analysing the implications of the results for management and policy makers, a number of shortcomings of applying a cross-sectional approach to an industry characterised by significant, lumpy and risky investments are identified and the potential benefits of a dynamic analysis, based on panel data, are enumerated.